

AAAI Summer Symposium 2025 on

AI-Driven Resilience: Building Robust, Adaptive Technologies for a Dynamic World

Multi-Scale Unrectified Push-Pull with Channel Attention for Enhanced Corruption Robustness

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USD AI Research Lab

Motivation

- Designing ML models that are robust to visible corruptions
 - Standard architectures such as VGG, Resnet e.t.c give good results on standard benchmarks like ImageNet, CIFAR
 - But **minor visible corruptions** in the data causes sharp performance loss

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We focus on Architectures that can generalize for **unseen corruptions**

Benchmark to test Image Corruptions

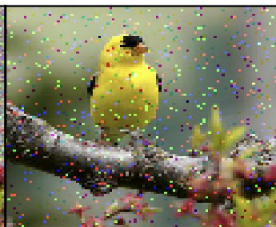
Gaussian Noise



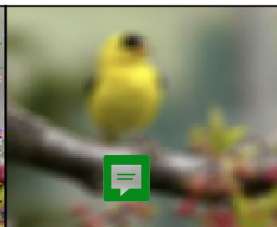
Shot Noise



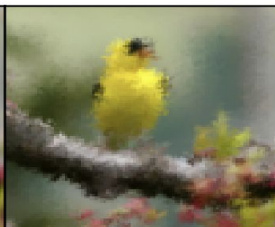
Impulse Noise



Defocus Blur



Frosted Glass Blur



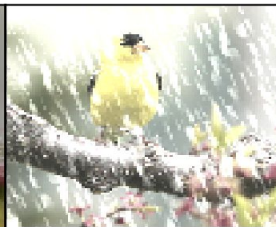
Motion Blur



Zoom Blur



Snow



Frost



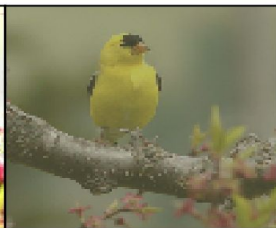
Fog



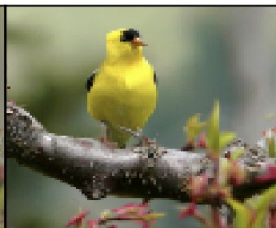
Brightness



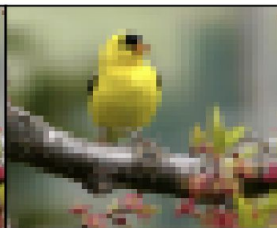
Contrast



Elastic



Pixelate

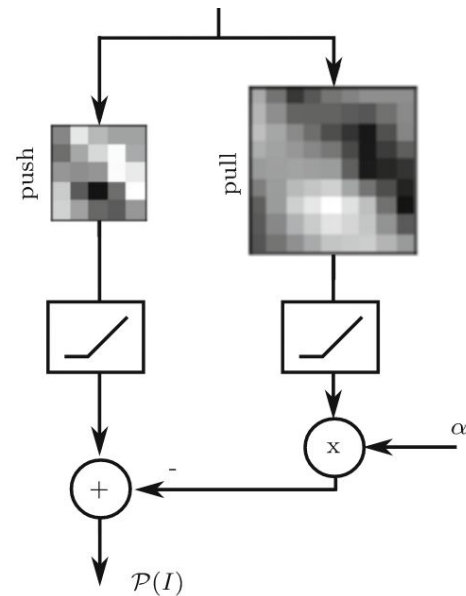


JPEG



Related Works : Push-Pull CNN Architecture

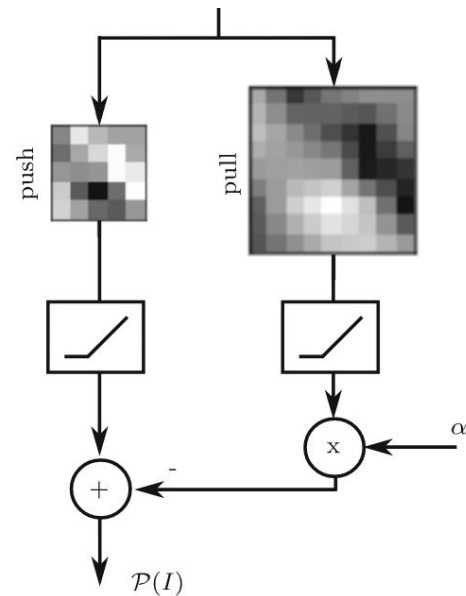
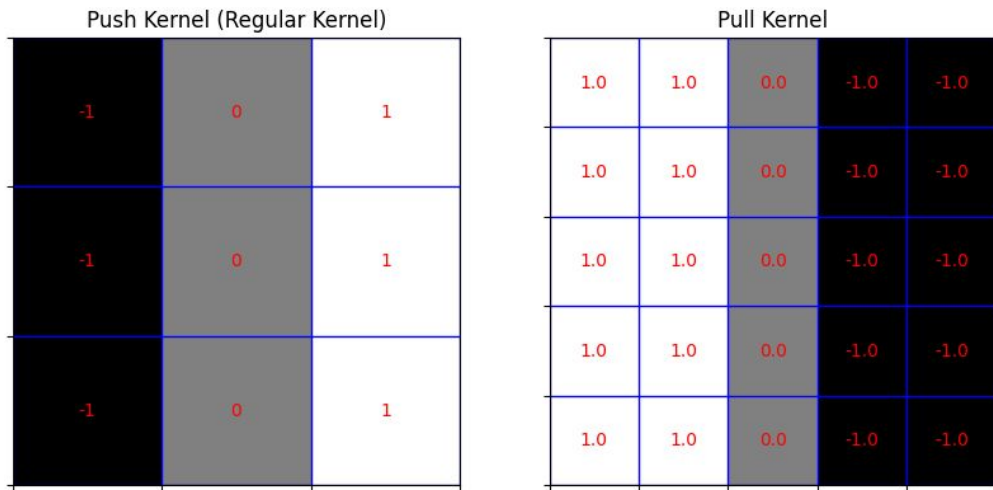
- A Modified CNN that computes response as the combination of two half-wave rectified convolutions, with kernels of different scales and opposite polarity.
- A biologically motivated model of certain neurons in the visual system that exhibit response suppression, known as push–pull inhibition.
- Noisy activation is suppressed on subtraction of Push and pull responses. Suppression is Due to ReLU Applied on activations (allowing only positive values to pass)



Enhanced robustness of convolutional networks with a push–pull inhibition layer

Related Works : Push-Pull CNN Architecture

Assume a Vertical Edge Detection Kernel. **Pull kernel** is scaled version of **Push kernel**, with opposite polarity

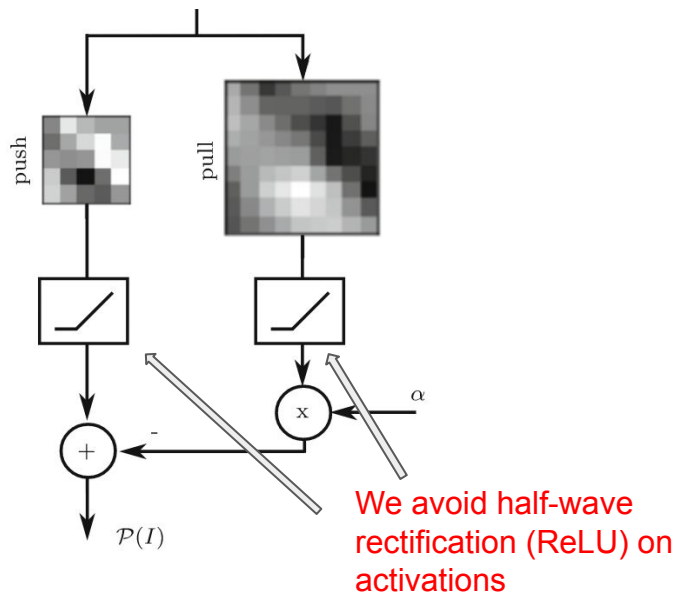


Enhanced robustness of convolutional networks with a push-pull inhibition layer

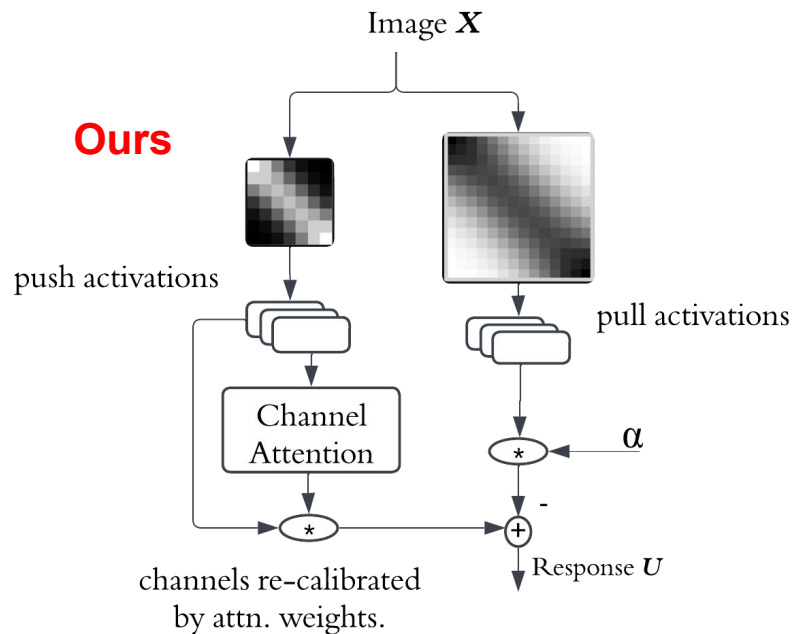
Our Contributions

Removing the half-wave rectification constraint to enable more expressive interactions between push and pull signals

Baseline Push-Pull



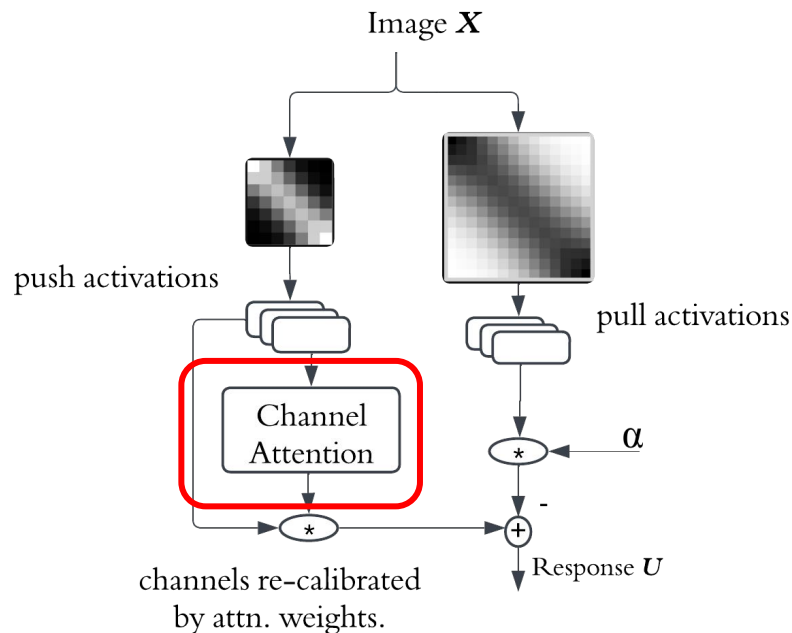
Ours



Enhanced robustness of convolutional networks with a push-pull inhibition layer

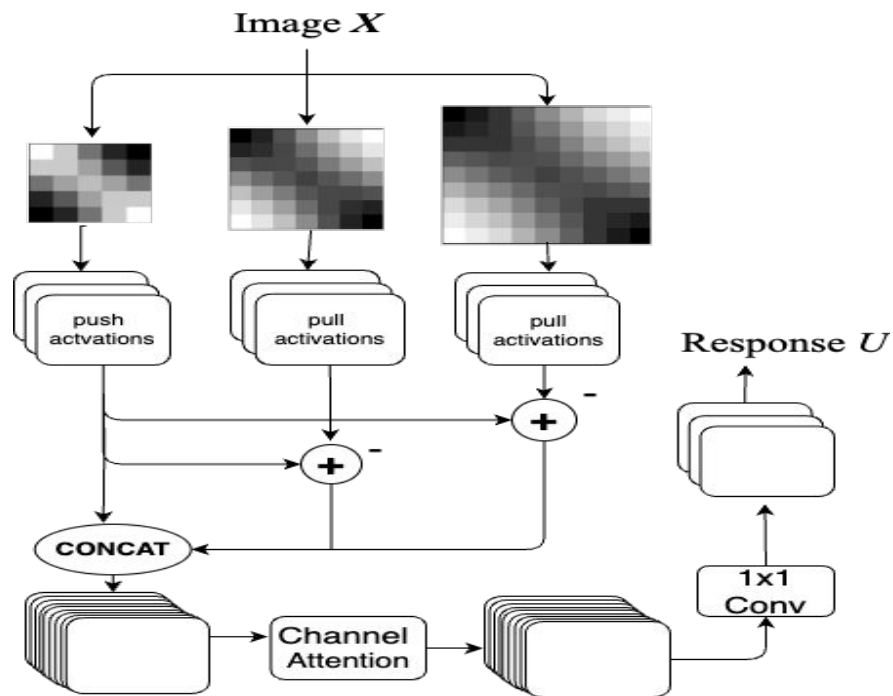
Our Contributions

Dynamic Attention mechanism to recalibrate feature responses in a way that, *amplifies discriminative channels* and *suppresses noise-dominated channels*

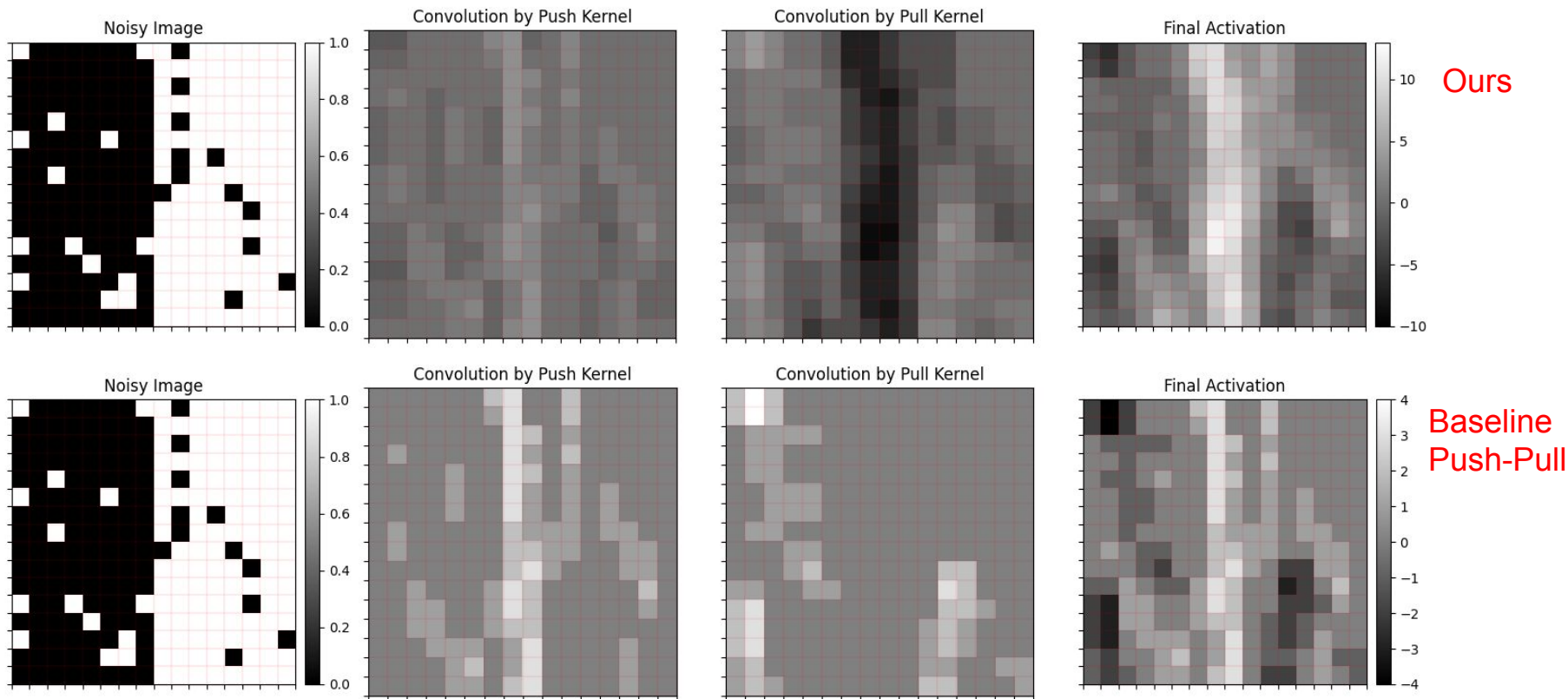


Our Contributions

Multi-scale push-pull (MSPP) framework that searches for pattern consistency across multiple spatial resolutions, reinforcing the model's ability to generalize under corruptions at varying scales.

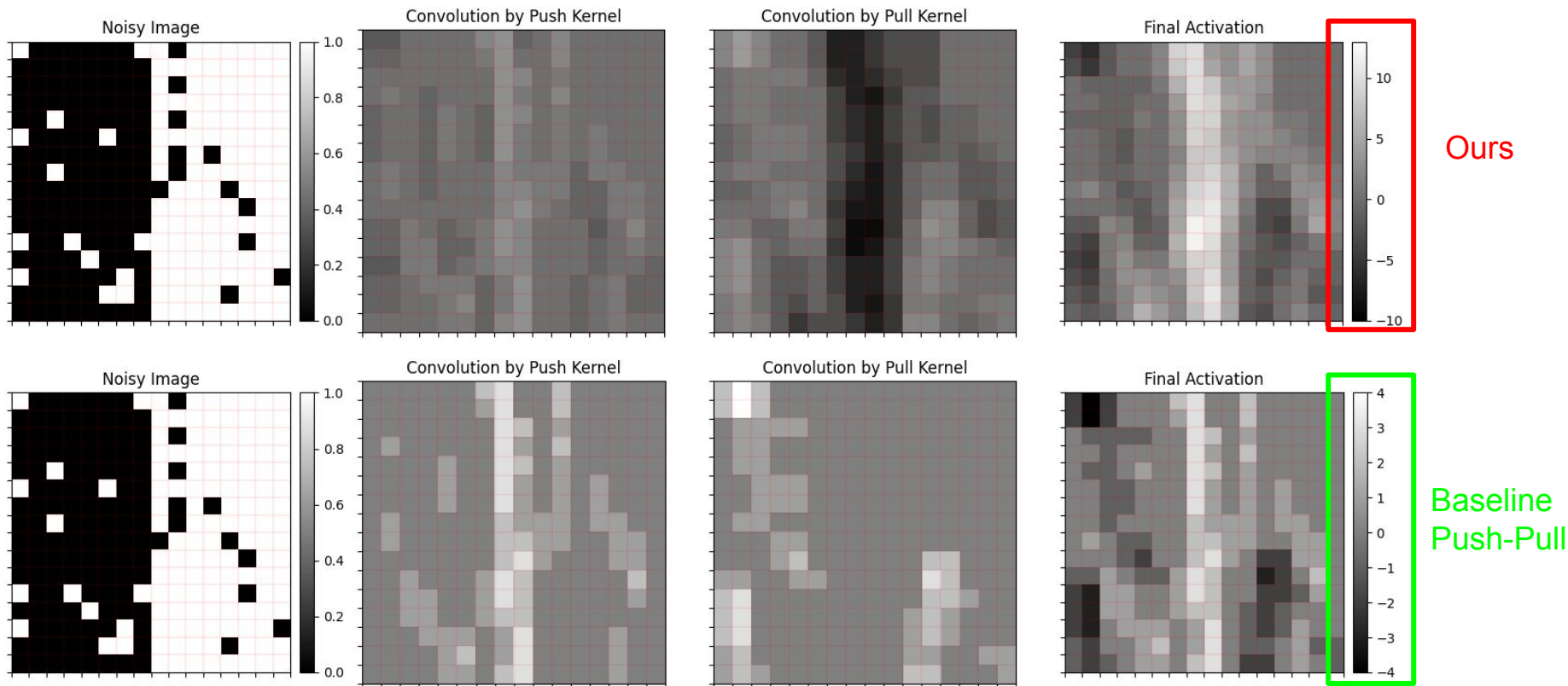


1. Remove Rectification in Push-Pull



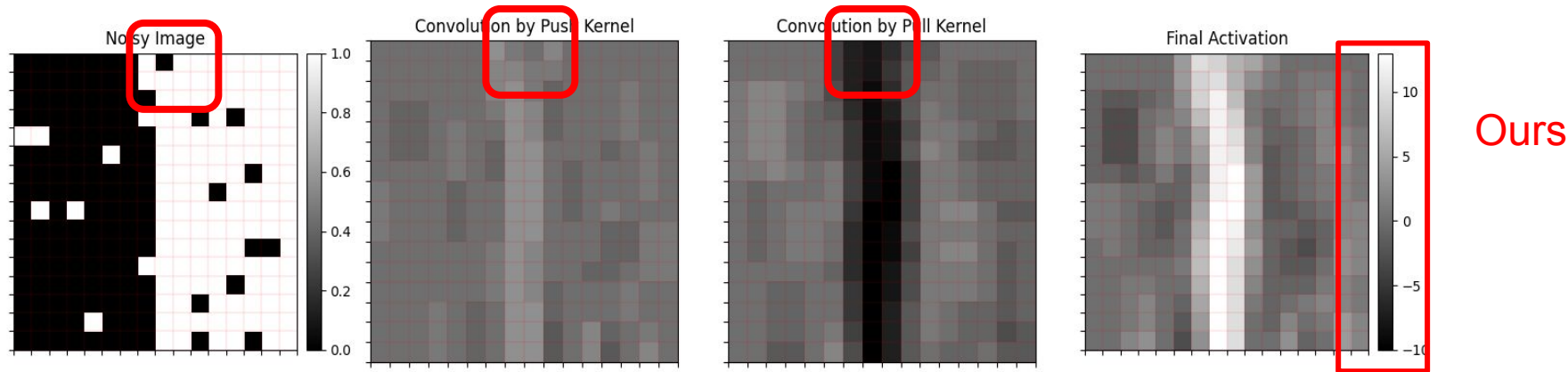
1. Remove Rectification in Push-Pull

Richer Feature Range, Prominent Features



1. Remove Rectification in Push-Pull

Feature Sharpening Effect

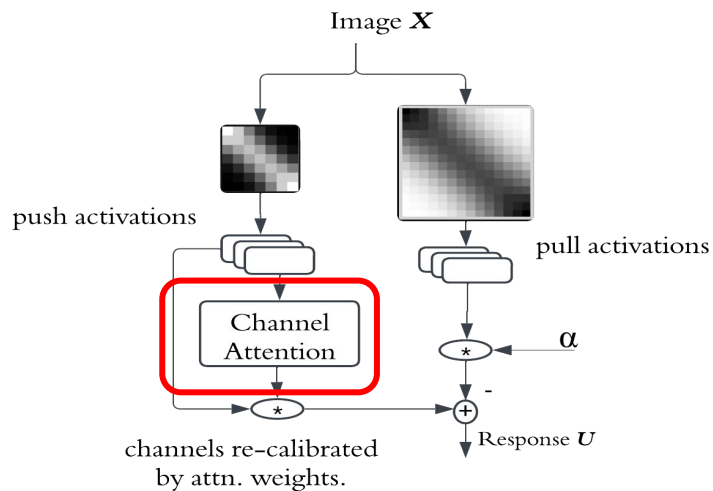
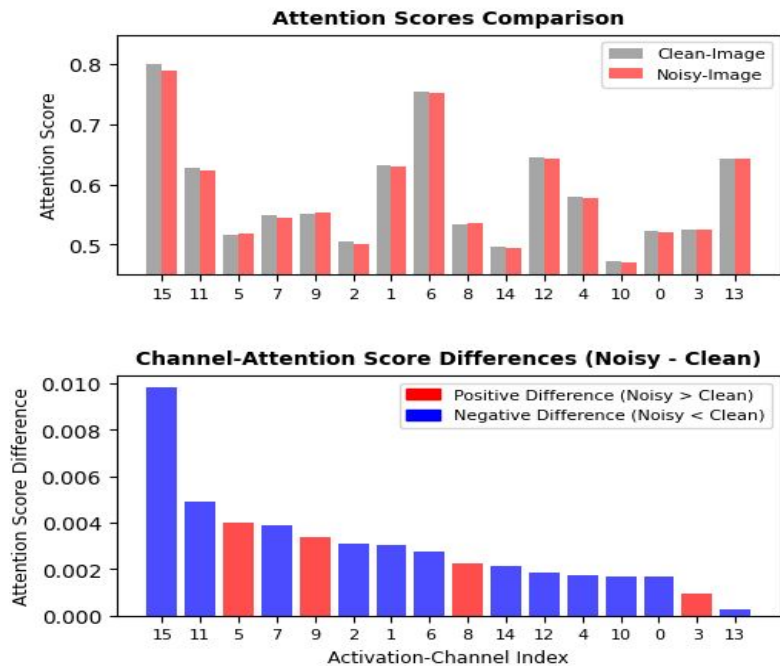


An Alternative Perspective :

- **Sharpens the activations strongly, if pattern is consistent over a larger region.**
- Even when the **smaller Push kernel can't detect feature due to local corruption**, Pull kernel can still detect due to its wider spatial dimension.

2. Channel Attention

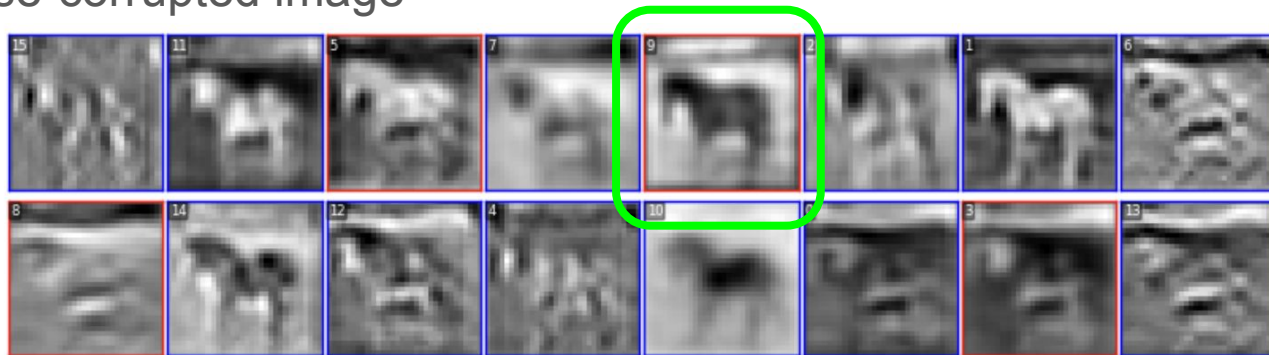
- Integrating Channel Attention alone improves on noise-robustness of Push-Pull Architecture. Attention shifts focus more to the channels with “visibly-discriminatory” features



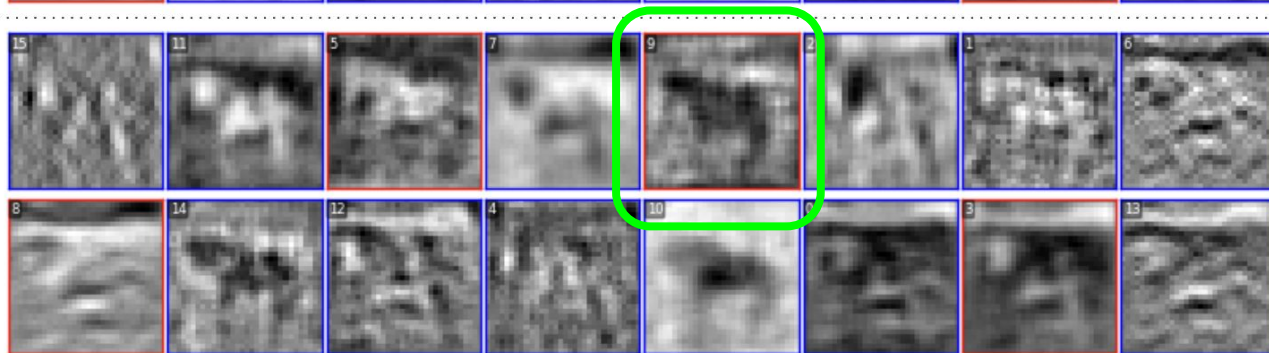
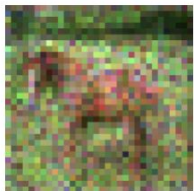
2. Channel Attention (Analysis)

Attention **Increases** Focus on **Channel 9**, where **Distinctive Feature** is preserved for noise-corrupted image

PP Attn activations
(clean image)



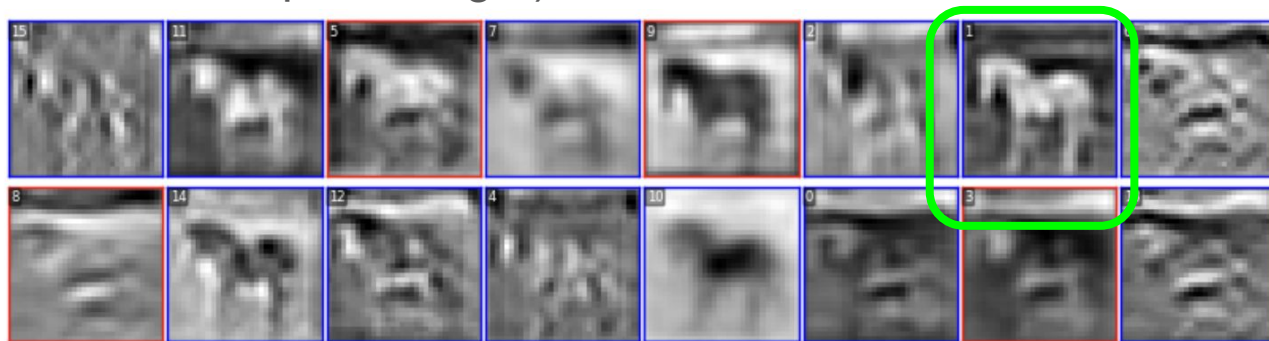
PP Attn activations
(speckle-noise)



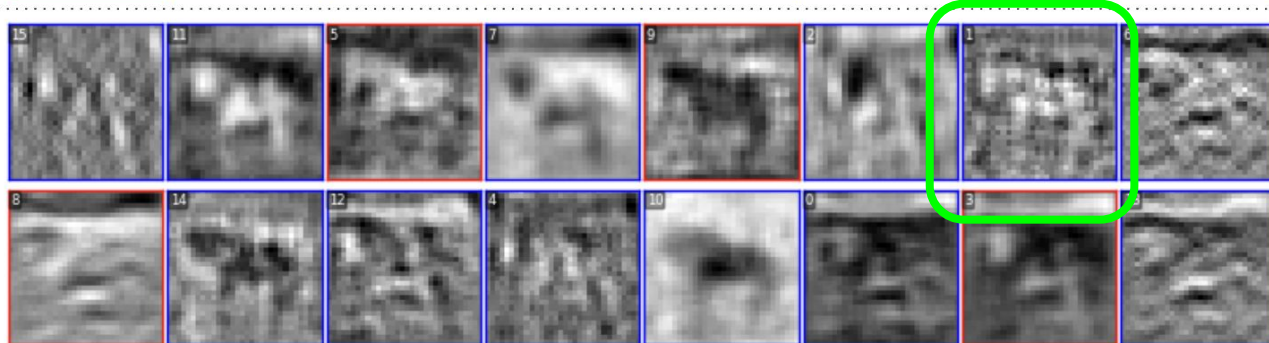
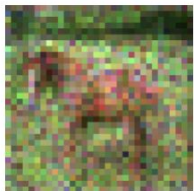
2. Channel Attention (Analysis)

Attention **Decreases** Focus from **Channel 1**, where **Distinctive Feature** is **NOT** preserved for noise-corrupted image)

PP Attn activations
(clean image)

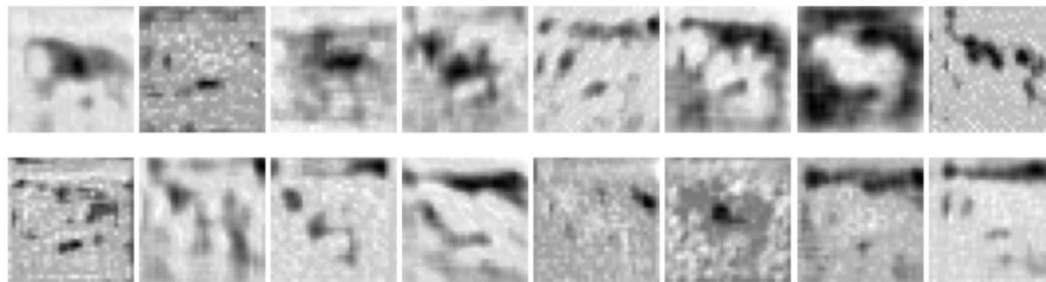


PP Attn activations
(speckle-noise)

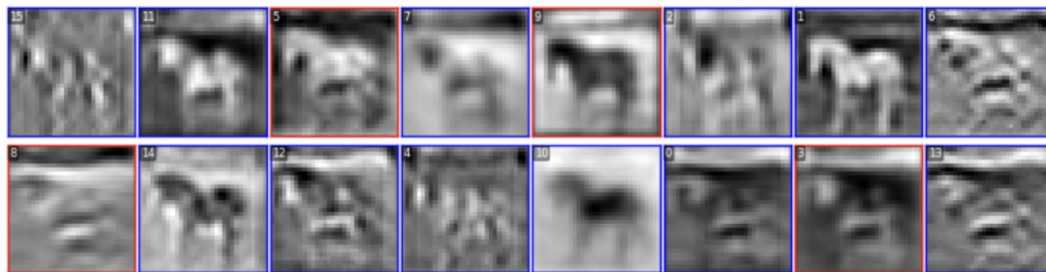


2. Channel Attention (Response maps with and without Attention)

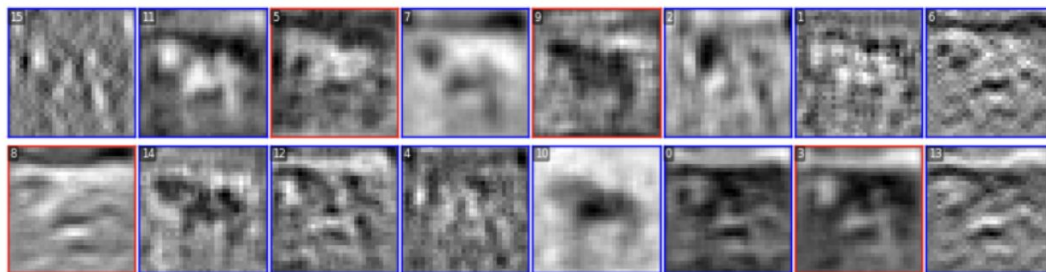
a) *PP ReLU* activations
(speckle-noise)



b) *PP Attn* activations
(clean image)

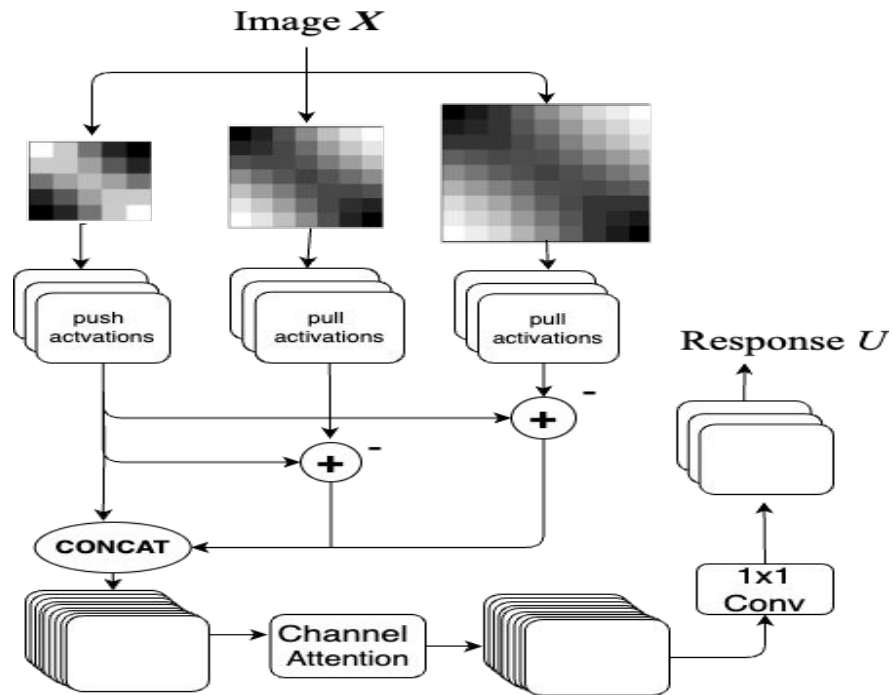


c) *PP Attn* activations
(speckle-noise)



3. Towards Multiple Pull Kernels

- Further complements the **Sharpening effect** of our *Modified Push-Pull*, if same “Pattern” is detected over an larger spatial area.
- Enables the push-pull layer to consider for corruptions that can occur at different spatial scales
- Response maps are concatenated followed by 1×1 Convolution to emulate response of a Normal CNN layer, while also reducing parameter-complexity.



Experimental Setup

- Resnet-20 and Densenet-40
- Swap Top **N** CNN Layers with our custom CNN Layers
- Train on CIFAR-10 Benchmark, with Minimal Data Augmentation
(Random Cropping & Horizontal Flipping)
- Test on CIFAR-10-C Benchmark (19 Visible Corruption Types) for Robustness

Results and Discussion

Noise Type	ResNet-20					DenseNet-40				
	PP ReLU (base-line)	PP + GELU (new base-line)	PP attn (ours)	PP attn + GELU (ours)	Mult. Scale PP attn + GELU (ours)	PP ReLU (base-line)	PP + GELU (new base-line)	PP attn (ours)	PP attn + GELU (ours)	Mult. Scale PP attn + GELU (ours)
Clean CIFAR-10	82.05	82.63	79.51	80.00	83.66	79.52	78.62	74.23	75.2	<u>79.04</u>
Gaussian Noise	57.06	62.9	65.90	<u>63.28</u>	56.85	52.15	56.36	<u>62.26</u>	64.12	56.98
Shot Noise	62.83	67.82	69.58	<u>67.48</u>	63.90	57.86	61.54	<u>66.02</u>	67.32	62.87
Impulse Noise	57.99	62.92	<u>59.65</u>	57.02	57.61	56.38	57.15	57.28	58.41	<u>58.33</u>
Speckle Noise	63.84	68.33	69.97	67.74	64.96	58.90	61.91	<u>65.86</u>	67.41	63.30
<i>Avg Noise Acc</i>	60.43	<u>65.49</u>	66.27	65.43	60.54	56.32	59.24	<u>62.85</u>	64.35	60.92
Gaussian Blur	67.75	69.03	<u>75.21</u>	76.05	73.32	62.29	62.70	<u>69.48</u>	70.08	63.63
Defocus Blur	72.62	73.53	77.06	<u>77.86</u>	76.82	67.84	67.72	<u>71.38</u>	71.92	68.50
Glass Blur	72.04	72.53	<u>75.77</u>	76.66	74.55	65.54	67.49	<u>71.37</u>	72.05	69.66
Motion Blur	67.99	68.61	<u>72.99</u>	73.43	71.71	61.97	62.69	<u>67.56</u>	68.22	63.74
Zoom Blur	69.10	70.86	<u>75.92</u>	76.42	74.48	63.58	63.91	<u>69.68</u>	70.51	65.24
<i>Avg Blur Acc</i>	69.9	70.91	75.39	71.34	<u>74.17</u>	64.66	64.9	<u>69.89</u>	70.55	66.15
Snow	72.77	72.25	73.15	<u>74.08</u>	75.34	67.47	<u>67.79</u>	67.44	67.48	69.23
Frost	70.73	70.45	71.18	<u>72.19</u>	73.95	64.43	<u>65.26</u>	64.46	63.60	66.12
Fog	<u>66.65</u>	64.88	62.06	63.91	68.16	62.09	57.77	55.08	55.91	<u>60.17</u>
Brightness	<u>79.41</u>	78.89	76.24	76.37	79.83	75.42	74.05	70.45	69.83	<u>74.19</u>
Contrast	<u>49.93</u>	49.74	46.88	49.17	51.32	46.49	43.62	42.17	43.28	<u>44.17</u>
Elastic Transform	75.03	75.4	75.87	<u>76.62</u>	77.49	70.77	70.09	70.83	71.63	<u>71.13</u>
Pixelate	80.6	<u>80.45</u>	78.82	79.65	81.81	74.71	<u>75.59</u>	73.69	74.61	77.05
JPEG Compression	<u>75.07</u>	76.79	73.35	73.49	76.08	71.68	72.65	69.29	69.37	<u>72.39</u>
Spatter	76.80	<u>77.11</u>	75.09	76.11	78.41	<u>73.60</u>	72.57	70.48	70.96	74.31
Saturate	72.39	<u>73.45</u>	68.94	68.78	72.96	69.20	66.96	61.10	61.45	<u>66.76</u>
Avg Corruption Acc	68.98	70.31	70.72	<u>70.86</u>	70.97	64.40	64.62	65.57	66.20	<u>65.67</u>

Table 1: Evaluating different push-pull layers on CIFAR-10 and CIFAR-10-C for ResNet-20 and DenseNet-40. For readability purpose, the best results are in **bold** and the second best results are underlined.

Results and Discussion

- **Removing Relu Constraint & Dynamic Attention Layer**, individually contributes towards improving the noise-robustness. Combination gives the best result.
- **Multi-Scale Push-pull** architecture gives best result on both **CIFAR-10 (clean)** and **CIFAR-10-C (noisy)** benchmark. However, with increased resource usage (parameters and computation) for each additional Push-kernel Introduced.
- These custom CNN's give best performance when swapped with regular CNN at top layers in standard architectures. Idea being, Once these custom layers catch essential features from raw noisy-inputs, regular CNN's will work fine.

Conclusion

- Un-rectification : Transformed Push-Pull from Noise-suppression to a Linear Activation Enhancement layer which promote patterns that's consistent across spatial scales.
- Channel Attention Mechanism: Adaptive focus on discriminative features under corruption.
- Multi-Scale Framework: Improved resilience by leveraging Pull Kernels of varying spatial scales. Effectively mitigated the robustness–accuracy trade-off, improving both clean and corrupted data performance.